

UNIT – 5

ASSESSMENT – 1

OUTPUT

USER-ITEM INTERACTIONS

('U1', 'I1', 5)

('U1', 'I2', 4)

('U1', 'I3', 2)

...

CONTENT-BASED RECOMMENDATIONS

['I1', 'I2', 'I4']

SIMILAR USERS

['U2', 'U3', 'U4']

COLLABORATIVE RECOMMENDATIONS

['I4', 'I5']

FINAL HYBRID RECOMMENDATIONS

I1

I2

I4

I5

INTELLIGENT AGENT

Agent detected positive feedback.

Agent increases priority of similar items.

1.

```
('ZoneA', 'Flood', 8)
('ZoneB', 'Earthquake', 7)
('ZoneC', 'Flood', 3)
('ZoneA', 'Flood', 9)
```

HIGH INTENSITY SENSOR EVENTS

```
-----
('ZoneA', 'Flood', 8)
('ZoneB', 'Earthquake', 7)
('ZoneA', 'Flood', 9)
```

SOCIAL MEDIA REPORT COUNT

```
-----
(('ZoneA', 'Flood'), 2)
(('ZoneB', 'Earthquake'), 1)
(('ZoneC', 'Flood'), 1)
```

FUSED EVIDENCE

```
-----
(('ZoneA', 'Flood'), 5)
(('ZoneB', 'Earthquake'), 3)
...
```

DISASTER ALERTS

```
-----
ALERT: ('ZoneA', 'Flood') | Evidence: 5
ALERT: ('ZoneB', 'Earthquake') | Evidence: 3
```

INTELLIGENT RESPONSE AGENT

```
-----
```

2. Agent response for ('ZoneA', 'Flood') : Iss  arly warning and recommend evacuation

TRANSACTION DATA

('T1', 'U1', 500, 'India')
('T2', 'U2', 1200, 'India')
('T3', 'U1', 15000, 'Unknown')
...

SUSPICIOUS TRANSACTIONS

('T3', 'U1', 15000, 'Unknown')
('T5', 'U2', 25000, 'Unknown')
('T7', 'U1', 18000, 'Unknown')

SUSPICIOUS TRANSACTIONS PER USER

('U1', 2)
('U2', 1)

FRAUD DETECTION AGENT

HIGH RISK USER: U1 -> Account requires investigation.
USER: U2 -> Transaction flagged for review.

3.

SUPPLY CHAIN ANALYSIS

Laptop | Stock: 20 | Demand: 35 | Shortage: 15
Phone | Stock: 50 | Demand: 30 | Shortage: 0
Tablet | Stock: 15 | Demand: 25 | Shortage: 10
Monitor | Stock: 8 | Demand: 20 | Shortage: 12
Keyboard | Stock: 40 | Demand: 15 | Shortage: 0

INVENTORY AGENT

Laptop -> Restock 15 units
Phone -> Inventory sufficient
Tablet -> Restock 10 units
Monitor -> Restock 12 units
Keyboard -> Inventory sufficient

4.

5.

```
STUDENT LEARNING DATA
-----
('S1', 'Python', 85, 8)
('S1', 'Math', 55, 4)
('S1', 'AI', 70, 6)
...

AVERAGE STUDENT SCORE
-----
S1 Average: 70.0
S2 Average: 76.67
S3 Average: 53.33

WEAK TOPICS
-----
S1 -> Math | Score: 55
S2 -> Python | Score: 60
S3 -> Python | Score: 45
S3 -> AI | Score: 50

LEARNING AGENT RECOMMENDATIONS
-----
S1 | Weak Subject: Math | Recommendation: Revision videos + practice exercises
S3 | Weak Subject: Python | Recommendation: Beginner lessons + practice quizzes

AGENT FEEDBACK ADAPTATION
-----
S1 -> Continue current learning path
S2 -> Continue current learning path
S3 -> Agent recommends additional support
```